Automated Investment Analytics & Attribution System

1. Executive Summary

This project delivers a high-fidelity, automated analytical framework that bridges raw financial data with actionable investment intelligence. By integrating **Python**, **PostgreSQL**, and **Tableau**, the system provides real-time tracking of portfolio performance, dynamic **Brinson-Fachler attribution**, and comprehensive peer benchmarking. The final output enables Portfolio Managers to identify Alpha drivers and communicate complex risk metrics to stakeholders with clarity and precision.

2. Technical Architecture & Data Pipeline

To ensure the "Automation Integration" requirement, a three-tier architecture was implemented:

- **Data Ingestion (Python)**: Utilized yfinance to automate the extraction of daily returns for portfolio holdings and peer benchmarks (QQQ, IVV).
- **Database Management (PostgreSQL)**: Engineered a relational schema to store market data, dimensions, and holdings. Implemented automated "Clean & Load" logic using SQLAlchemy to ensure data integrity.
- **Analytics Layer (SQL Views)**: Developed advanced SQL views to perform complex calculations server-side, reducing Tableau's processing load and ensuring a "Single Source of Truth".

3. Quantitative Methodology

A. Brinson-Fachler Attribution

The core of the "Driving Factor Analysis" is the Brinson-Fachler model, which decomposes excess returns into two primary effects:

- **Selection Effect**: Measures the value added by picking specific stocks within a sector (e.g., Apple vs. the Tech sector).
- **Allocation Effect**: Measures the impact of overweighting or underweighting specific sectors relative to the benchmark.

B. Risk Performance Metrics

The system automatically computes rolling risk-adjusted metrics:

- **Sharpe Ratio**: Evaluates excess return per unit of volatility.

- **Alpha & Beta**: Quantifies market sensitivity and idiosyncratic outperformance.
- **Rolling Volatility**: Tracks risk fluctuations over a 252-day window to monitor stability.

4. Visualization & Insight Delivery

The **Tableau Dashboard** was designed for "Executive Communication":

- **Attribution Waterfall**: A custom Gantt-style chart that visualizes how each sector contributed to the total Alpha, highlighting the **Technology sector** as a key driver.
- **Peer Analysis**: A multi-axis time-series chart comparing the portfolio's net asset value (NAV) against **SPY, QQQ, and IVV**.
- **Interactive Storytelling**: Includes an English Executive Summary that translates quantitative indicators into strategic business narratives.

5. Key Results

- **Zero-Manual Effort**: Eliminated the need for manual spreadsheet updates through a fully automated ETL pipeline.
- **Enhanced Decision Support**: Provided a granular view of sector-level performance, allowing for rapid rebalancing based on Selection Alpha.
- **Scalability**: The modular SQL/Python backend allows for the effortless addition of new asset classes or alternative benchmarks.